

CODE FOR THE RELAY MODULE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin connections
const int moisturePin = 32; // AO of soil sensor
const int valvePin = 27;    // Relay for valve
const int pumpPin = 26;     // Relay for pump

// Calibration (tune these for your sensor)
const int dryValue = 4000;  // ADC when dry (in air)
const int wetValue = 1000;  // ADC when wet (in water)

// Control thresholds
const int moistureThreshold = 35; // Start watering if below this

bool watering = false; // Track watering state

void setup() {
  lcd.init();
  lcd.backlight();

  pinMode(moisturePin, INPUT);
  pinMode(valvePin, OUTPUT);
  pinMode(pumpPin, OUTPUT);

  // Make sure both are off initially
  digitalWrite(valvePin, LOW);
  digitalWrite(pumpPin, LOW);

  lcd.setCursor(0, 0);
  lcd.print("Smart Water Sys");
  lcd.setCursor(0, 1);
  lcd.print("Initializing...");
  delay(2000);
  lcd.clear();
}

void loop() {
  int analogValue = analogRead(moisturePin);

  // Convert ADC to % moisture
```

```

int moisturePercent = map(analogValue, dryValue, wetValue, 0, 100);
moisturePercent = constrain(moisturePercent, 0, 100);

// Show moisture on LCD
lcd.setCursor(0, 0);
lcd.print("Moisture: ");
lcd.print(moisturePercent);
lcd.print("% ");

// Control logic
if (moisturePercent < moistureThreshold && !watering) {
    watering = true;

    // Step 1: Turn on valve
    digitalWrite(valvePin, LOW);
    lcd.setCursor(0, 1);
    lcd.print("Valve: ON Pump:OFF");
    delay(2000);

    // Step 2: Turn on pump
    digitalWrite(pumpPin, LOW);
    lcd.setCursor(0, 1);
    lcd.print("Pump:ON ");
}
else if (moisturePercent >= (moistureThreshold + 5) && watering) {
    // Step 1: Turn off pump
    digitalWrite(pumpPin, HIGH);
    lcd.setCursor(0, 1);
    lcd.print("Valve: ON Pump:OFF");
    delay(2000);

    // Step 2: Turn off valve
    digitalWrite(valvePin, HIGH);
    lcd.setCursor(0, 1);
    lcd.print("Pump:OFF ");
    watering = false;
}

delay(1000);
}
void setup() {
    pinMode(26, OUTPUT);
}

```

```
void loop() {  
  digitalWrite(26, LOW);  
  delay(5000);  
  digitalWrite(26, HIGH);  
  delay(100);  
}
```

CODE FOR THE SOIL MOISTURE SENSOR AND LCD

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2); // I2C address and LCD size
```

```
const int analogPin = 32; // AO pin of HW-080 to ESP32 GPIO 32
```

```
int analogValue = 0;
```

```
// Calibration values (adjust after testing)
```

```
const int dryValue = 3500; // Sensor reading in air
```

```
const int wetValue = 1500; // Sensor reading fully in water
```

```
void setup() {
```

```
  lcd.init();
```

```
  lcd.backlight();
```

```
  pinMode(analogPin, INPUT);
```

```
  lcd.setCursor(0, 0);
```

```
  lcd.print("Soil Moisture");
```

```
  delay(2000);
```

```
  lcd.clear();
```

```
}
```

```
void loop() {
```

```
  analogValue = analogRead(analogPin);
```

```
  // Convert ADC to percentage (100% = wet, 0% = dry)
```

```
  int moisturePercent = map(analogValue, dryValue, wetValue, 0, 100);
```

```
  moisturePercent = constrain(moisturePercent, 0, 100);
```

```
  lcd.setCursor(0, 0);
```

```
  lcd.print("Moisture: ");
```

```
  lcd.print(moisturePercent);
```

```
  lcd.print("% ");
```

```
  delay(1000);
```

```
}
```